

3S25:  
Introduction to  
Fourier Analysis

Sam Condon

Linear  
Time-Invariant  
Systems

General Fourier  
Series  
Coefficients

Fourier Series of  
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# **SAGI Summer School 2025: Introduction to Fourier Analysis**

Sam Condon

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# Linear Time-Invariant Systems

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- ① In the previous lecture, we started with a simple RC circuit and wrote the differential equation which describes its behavior:

$$x(t) = RC \frac{dy(t)}{dt} + y(t) \quad (1)$$

- ② Next, we saw that if we take the input  $x(t) = e^{i\omega t}$ , we get for the output:

$$y(t) = H(\omega)e^{i\omega t} \quad (2)$$

where  $H(\omega)$  is the RC circuit **transfer function**.

- ③ For the RC circuit, we have:

$$H(\omega) = \frac{1}{1 + i\omega RC} \quad (3)$$

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- ① Further, we saw that for an arbitrary linear combination of complex exponentials, we have the following:

$$x(t) = \sum_{n=-\infty}^{\infty} e^{i\omega_n t} \longrightarrow y(t) = \sum_{n=-\infty}^{\infty} H(\omega_n) a_n e^{i\omega_n t} \quad (4)$$

- ② In the physical world, we don't have complex exponentials, so we take the real part of both sides to represent a physical signal:

$$x(t) = \sum_{n=-\infty}^{\infty} \cos(\omega_n t) \longrightarrow y(t) = \sum_{n=-\infty}^{\infty} |H(\omega_n)| a_n \cos(\omega_n t + \angle H(\omega_n)) \quad (5)$$

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- ① Where for the RC circuit:

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad (6)$$

$$\angle H(\omega) = \tan^{-1}(-\omega RC) \quad (7)$$

- ② For the RC circuit, when we input an arbitrary linear combination of sinusoids, the output is the **same linear combination** of sinusoids, with each sinusoid scaled by the factor  $|H(\omega_n)|$  and phase shifted by  $\angle H(\omega_n)$ .
- ③ In fact, this principal is much more general and applies to many more systems than just the RC circuit.
- ④ Let's define the systems for which this principal applies.

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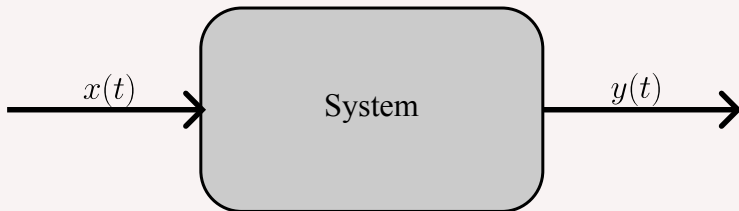
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## Definition: System

A **system** is any physical entity which transforms an input signal  $x(t)$  to an output signal  $y(t)$ .



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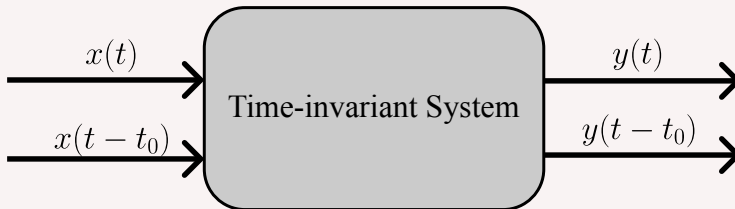
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## Definition: Time-invariant System

A system is **time-invariant** if for any input/output pair  $x(t) \rightarrow y(t)$  and any value  $t_0$ , we have the input/output pair  $x(t - t_0) \rightarrow y(t - t_0)$ :



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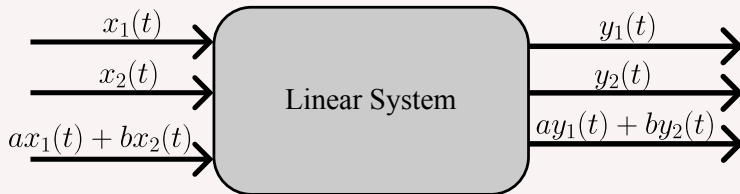
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## Definition: Linear System

A system is linear, if for any two input output pairs  $x_1(t) \rightarrow y_1(t)$ ,  $x_2 \rightarrow y_2(t)$ , we have the input/output pair  $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$ , where  $a$  and  $b$  are any scalar functions:



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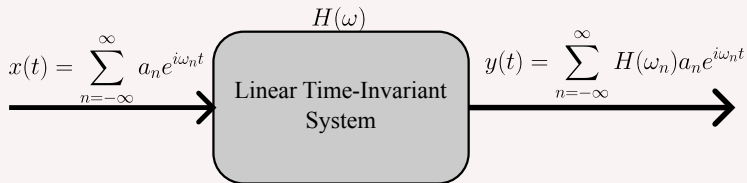
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## Definition: Linear Time Invariant System

Finally, a **Linear-Time Invariant (LTI) system** is a system that is both linear and time-invariant. It turns out that for any LTI system, we have the input/output pair shown in the diagram below:



The input/output pairs described by equations 4 and 5 are true for all LTI systems.



# General Fourier Series Coefficients

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- ① This property of LTI systems is incredibly powerful. It turns out, that **nearly all periodic signals** can be written in the form given by eqn. (4).

## Definition: Periodic Signal

A signal  $x(t)$  is periodic if for every value  $t$ , there exists a non-zero value  $T$  such that:

$$x(t) = x(t \pm T) \quad (8)$$

The value  $T$  is called the period of  $x(t)$ , if it is the minimum nonzero value for which this property is satisfied. The frequency of  $x(t)$ ,  $f$  is related to the period with:

$$f = \frac{1}{T} \quad (9)$$

- ② **Aperiodic signals** (signals which are not periodic) can be written in a similar form, where the summation turns to an integral. We will come back to this idea soon!

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- ① The coefficients  $a_n$  in equation 4 are known as **Fourier Series Coefficients**.
- ② How can we find these coefficients for some periodic input signal  $x(t)$ ?
- ③ Start by writing equation 4 again here:

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{i\omega_n t} \quad (10)$$

- ④ Now suppose that the signal  $x(t)$  is periodic with period  $T$ . Because  $x(t)$  is periodic, every complex exponential term in equation 10 must have a frequency related to the period  $T$  by  $\omega_n = \frac{2\pi n}{T} = n\omega_0$  where  $n$  is an integer. The summation is then:

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{in\omega_0 t} \quad (11)$$

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- ❶ Multiplying both sides by  $e^{-ik\omega_0 t}$  where  $k$  is an integer, and integrating over one period:

$$\int_t^{t+T} x(t) e^{-ik\omega_0 t} dt = \int_t^{t+T} \sum_{n=-\infty}^{\infty} a_n e^{i(n-k)\omega_0 t} dt \quad (12)$$

$$= \sum_{n=-\infty}^{\infty} a_n \int_t^{t+T} e^{i(n-k)\omega_0 t} dt \quad (13)$$

- ❷ When  $n = k$  we have  $e^{i(n-k)\omega_0 t} = 1$ . When  $n \neq k$ , the integral on the right hand side is zero. This is because we are integrating a cos and a sin over one period. The area under a cos and a sin function over one period is always zero.

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- ❶ Writing this out explicitly:

$$\int_t^{t+T} e^{j(n-k)\omega_0 t} dt = \begin{cases} 0, & n \neq k \\ T, & n = k \end{cases} \quad (14)$$

- ❷ Substituting this into equation 13, we get:

$$a_k = \frac{1}{T} \int_t^{t+T} x(t) e^{-ik\omega_0 t} dt \quad (15)$$

- ❸ Thus we have determined the fourier series coefficients for a general periodic signal  $x(t)$ !

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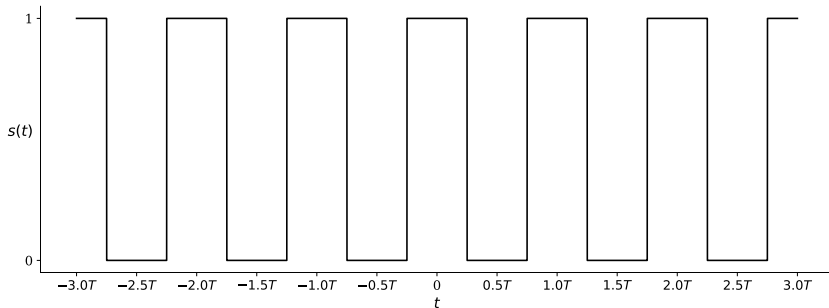
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- ① Let's apply this to find the Fourier Series coefficients for the square wave signal that we used as an input to the RC circuit earlier. We can define a square wave over one period  $T$  as:

$$s(t) = \begin{cases} 1, & |t| \leq \frac{T}{4} \\ 0, & \frac{T}{4} < |t| \leq \frac{T}{2} \end{cases} \quad (16)$$

- ② Several periods of this square wave are plotted below:



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- ① Now, we want to apply equation 15 to this signal. We need to integrate over one full period  $T$ , but we can choose  $t$  to be anything. Let's take  $t = -T/2$ :

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} s(t) e^{-ik\omega_0 t} dt \quad (17)$$

- ② Simplifying, using equation 16:

$$a_k = \frac{1}{T} \int_{-T/4}^{T/4} e^{-ik\omega_0 t} dt \quad (18)$$

$$= \left. \frac{e^{-ik\omega_0 t}}{-ik\omega_0 T} \right|_{-T/4}^{T/4} \quad (19)$$

$$= \frac{2}{k\omega_0 T} \left[ \frac{e^{ik\omega_0 T/4} - e^{-ik\omega_0 T/4}}{2i} \right] \quad (20)$$

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① Using  $\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$ :

$$a_k = \frac{2 \sin(k\omega_0 T/4)}{k\omega_0 T} \quad (21)$$

$$= \frac{2 \sin(k\pi/2)}{k\pi} \quad (22)$$

② This is only determined for  $k \neq 0$ . For  $k = 0$  we get:

$$a_0 = \frac{1}{T} \int_{-T/4}^{T/4} dt = \frac{1}{2} \quad (23)$$

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- ① Substituting the coefficients found in equations 22 and 23 into equation 11, we have the following Fourier Series representation of the square wave:

$$s(t) = \frac{1}{2} + \sum_{n=-\infty}^{-1} \frac{2 \sin(n\pi/2)}{n\pi} e^{in\omega_0 t} + \sum_{n=1}^{\infty} \frac{2 \sin(n\pi/2)}{n\pi} e^{in\omega_0 t} \quad (24)$$

- ② Now, because  $\sin(-x) = -\sin(x)$ , we can combine the two summation terms:

$$s(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2 \sin(n\pi/2)}{n\pi} (e^{in\omega_0 t} + e^{-in\omega_0 t}) \quad (25)$$



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## Result: Square Wave Fourier Series Representation

Finally, we arrive at the compact Fourier Series representation for the full square wave:

$$s(t) = \frac{1}{2} + 2 \sum_{n=1}^{\infty} \frac{2 \sin(n\pi/2)}{n\pi} \cos(n\omega_0 t) \quad (26)$$